

Git & GitHub — Beginner's Guide

Based on freeCodeCamp.org crash course by Sumit Saha

What is Git?

Git is a **version control system** that runs locally on your computer. It tracks every change made to your files — what changed, when, and by whom. It lets you save multiple versions and roll back to any previous state at any time.

Created by **Linus Torvalds** (also the creator of Linux), Git is one of the most enduring tools in software development.

Git vs GitHub

Git = the tool (runs on your machine). **GitHub** = the cloud platform where you host and share Git repositories. Think of Git as the coffee and GitHub as the coffee shop.

Core Workflow

Working Directory → **Staging Area** (git add) → **Local Repo** (git commit) → **Remote Repo** (git push)

Essential Commands

Command	What it does
git init	Initialize a new repository
git clone <url>	Clone a remote repository locally
git status	Check state of working directory & staging area
git add .	Stage all changes for commit
git commit -m "msg"	Save staged changes to local repo
git log	View commit history
git branch	List / manage branches
git checkout <branch>	Switch to a branch
git merge <branch>	Merge another branch into current
git push	Upload local commits to remote
git pull	Fetch + merge remote changes
git fetch	Download remote changes (no merge)
git stash	Temporarily save uncommitted work
git revert <hash>	Create a new commit that undoes a commit
git reset	Undo commits (rewrites history)
git rebase <branch>	Reapply commits on top of another branch

Branches

Branches let you work on features in isolation without affecting the main codebase. Use **git merge** to combine branches. Merge creates an extra merge commit; **git rebase** gives a cleaner, linear history but should **not** be used on shared/public branches as it rewrites commit history.

Pull Requests (PR)

A pull request is a request to merge your branch into another (usually main). It allows teammates to review your changes before they are merged. Steps:

- Push your branch to GitHub.
- Go to the Pull Requests tab and click New Pull Request.
- Select base (e.g. main) and compare (your branch) branches.
- Add a title and description, then click Create Pull Request.
- After review, click Merge Pull Request.

Merge Conflicts

Conflicts occur when two branches edit the same lines. Git marks the conflicting sections in the file. Edit the file to keep the correct version, then stage and commit to complete the merge.